pynasonde — Ionosonde Data Analysis in Python

A Hands-On Workshop for Graduate Students

Ionospheric Remote Sensing Group

VIPIR & DIGISONDE Analysis Toolkit

July 21, 2026

`pip install pynasonde` — github.com/shibaji7/pynasonde

Outline

1. Background
2. Installation & Quick Start
3. DIGISONDE Data
4. VIPIR Data
5. Ionospheric Analysis
6. Es Layer Imaging (In Development)
7. Hands-On Exercises
8. Summary

What is an Ionosonde?

- A **pulsed HF radar** ($\sim$1–30 MHz) transmitting vertically
- Each frequency reflects at the height where $f_p(h) = f$
- Records: virtual height $h'(f)$ vs frequency — the **ionogram**
- **Two instruments** in this toolkit:
  - **VIPIR** — 8-channel digital receiver, raw IQ storage
  - **DIGISONDE DPS4D** — SAO/RSF/DFT/DVL file formats

Key quantities

f	sounding frequency
$h'(f)$	virtual height (km)
f_p	plasma frequency
$N(h)$	electron density

Two Instrument Families

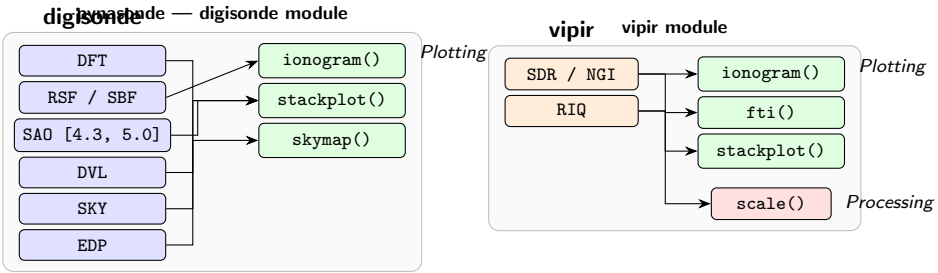
VIPIR

- Grubb et al. (2011)
- Raw IQ binary (`.RIQ`)
- 8 Rx channels, pulse-compressed
- Gate spacing: 1.499 km (WI937)
- 4 pulses/frequency (typical)
- Full analysis: Es imaging, N(h), absorption

DIGISONDE DPS4D

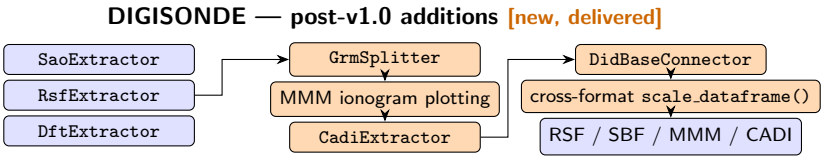
- Reinisch et al. (1997)
- Scaled data: `.SAO`
- Raw soundings: `.RSF`, `.SBF`
- Doppler spectra: `.DFT`
- Drift velocities: `.DVL`
- Sky maps: `.SKY`

pynasonde v1.0 Architecture (Chakraborty et al., SoftwareX 2026)



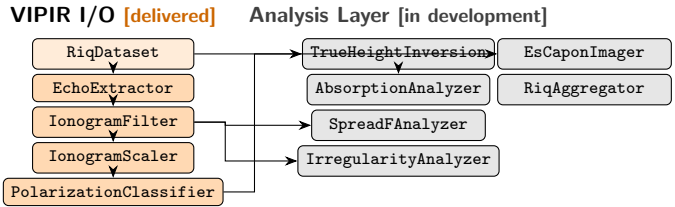
Adapted from Fig. 2, Chakraborty et al. (2026), *SoftwareX* 34, 102617.

Post-v1.0 Extensions — Digisonde I/O & Scaling



GRM auto-splits into RSF/SBF/MMM; MMM plotting, CADI echo extraction, DIDBase access, and cross-format scaling are new since v1.0 and are delivered, tested features of this release.

Post-v1.0 Extensions — VIPIR Analysis Layer (In Development)



Orange = new, delivered VIPIR I/O (echo extraction, filtering, scaling, polarization). Gray = analysis-layer modules that exist in the codebase but are **not yet validated/released** — shown here as a preview of ongoing work, not a claim of a stable API.

Installation

Terminal

```
# Option 1: PyPI
pip install pynasonde

# Option 2: developer install (recommended for this workshop)
git clone https://github.com/shibaji7/pynasonde.git
cd pynasonde
pip install -e ".[dev]"
```

Requirements

Python $\geq$ 3.11 — numpy, pandas, scipy, matplotlib, loguru

SAO — Electron Density Profiles

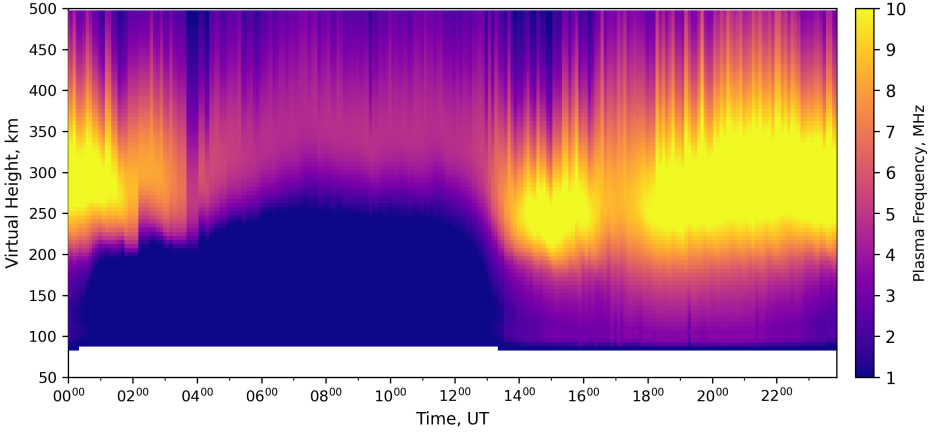
The `.SAO` file contains scaled ionogram parameters and height profiles:

```
from pynasonde.digisonde.parsers.sao import SaoExtractor

# Load a day of SAO files in parallel (n_procs workers)
df = SaoExtractor.load_SAO_files(
    folders=["data/KR835/"],
    func_name="height_profile",
    n_procs=4,
)
print(df.columns.tolist())
# ['datetime', 'height_km', 'plasma_freq_mhz', 'electron_density_cm3',
    ...]
```

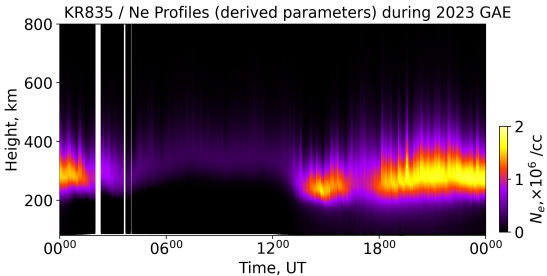
- `func_name="height_profile"` — N(h) profile per record
- `func_name="scaled_params"` — foF2, hmF2, MUF, foE

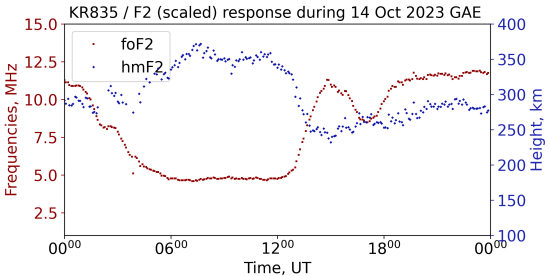
SAO — Isodensity Contour Plot



Daily isodensity contour from DPS4D SAO files at KR835. Each horizontal slice is one N(h) profile; colour encodes electron density.

SAO — N(h) Profile and F2 Parameters





Left: stacked N(h) profiles. Right: daily foF2 and hmF2 time series.

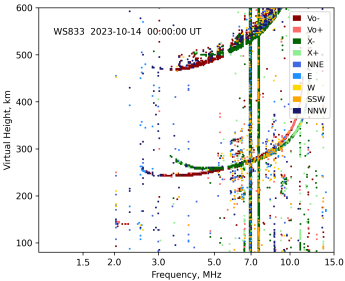
RSF — Direction-Coded Ionogram

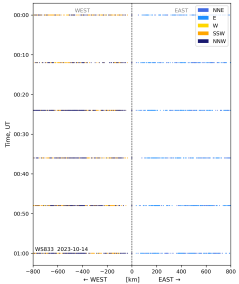
.RSF files store raw amplitude+phase for each sounding:

```
from pynasonde.digisonde.parsers.rsf import RsfExtractor

extractor = RsfExtractor("data/KR835_20230601.RSF")
extractor.extract()
df = extractor.to_dataframe()
extractor.plot_direction_ionogram()    # direction-coded ionogram
extractor.plot_directogram()           # full-day directogram
```

RSF — Direction-Coded Ionogram & Directogram





Left: direction-coded ionogram (colour = arrival azimuth). Right: daily directogram (time vs. ground distance).

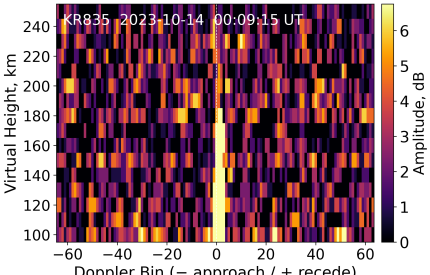
DFT — Doppler Waterfall & Spectra

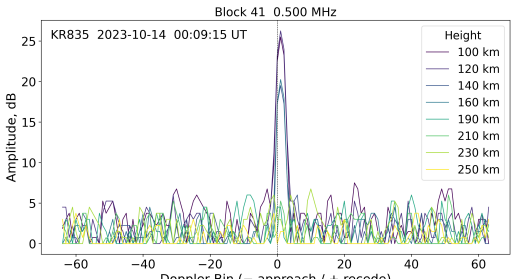
`.DFT` files contain per-height Doppler spectra:

```
from pynasonde.digisonde.parsers.dft import DftExtractor

dft = DftExtractor("data/KR835_20230601.DFT")
dft.extract()
dft.plot_doppler_waterfall()    # time-height Doppler waterfall
dft.plot_doppler_spectra()      # per-height spectra
```

DFT — Doppler Waterfall & Spectra





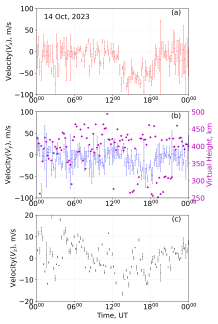
Left: Doppler velocity waterfall (height vs. time). Right: per-height Doppler spectra at one epoch.

DVL — Drift Velocity Stack Plot

.DVL files contain the ionospheric drift velocity estimates:

```
from pynasonde.digisonde.parsers.dvl import DvlExtractor

dvl = DvlExtractor.load_files(folder="data/KR835/", n_procs=4)
dvl.plot_stack()    # 3-panel Vx, Vy, Vz vs. time + virtual height overlay
```



GRM, RSF & MMM Ionogram Plotting

A GRM archive from GIRO DIDBase is a raw concatenation of RSF/SBF/MMM blocks; `GrmSplitter` auto-detects and routes each to the right parser:

```
from pynasonde.digisonde.parsers.grm import GrmSplitter
from pynasonde.digisonde.digi_plots import RsfIonogram

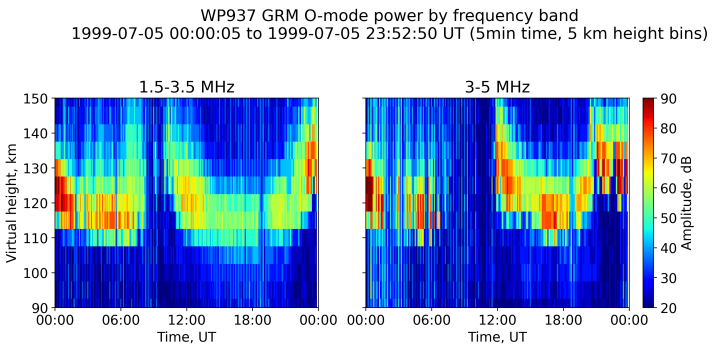
spl = GrmSplitter("AU930_20171470000.GRM")
print(spl.fmt)                # auto-detected: "RSF" | "SBF" | "MMM"
df = spl.load_dataframes()    # parallel-safe

r = RsfIonogram(figsize=(6, 5))
r.add_direction_ionogram(df)    # same plotting path as native RSF files
```

- MMM ionograms render as pcolormesh (combined O+X, and side-by-side O/X)
- GRM record type (RSF=7, SBF=3, MMM=9) is read directly from the archive

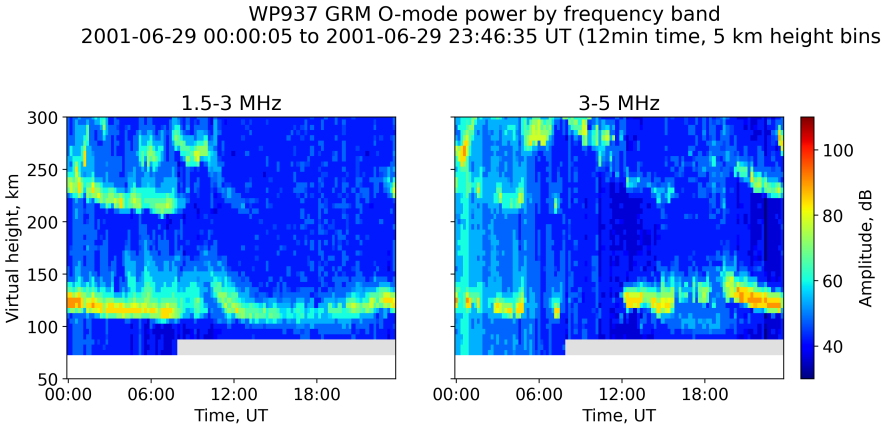
Special Analysis — Long-Term O-Mode Power Directogram (RSF/MMM)

GRM archives let decades-old WP937 soundings (RSF- and MMM-era DPS hardware alike) be binned onto a common, d256-compatible virtual-height grid (5 km bins) and a regular time grid, then split by frequency band to reveal diurnal E/Es persistence across a full day:



WP937, 1999-07-05: O-mode power (dB) vs. virtual height and UT, 1.5–3.5 and 3–5 MHz bands, 5-min × 5-km bins.

Special Analysis — Cross-Era Comparison (1999 vs. 2001)

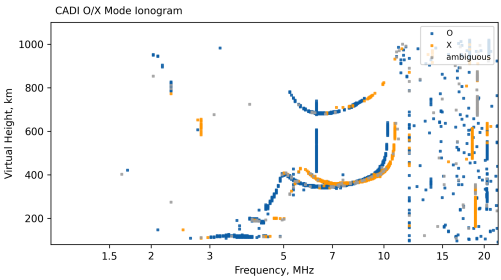


WP937, 2001-06-29: same binning scheme, 12-min cadence (lower duty cycle that day). The persistent 110–140 km E/Es layer and its diurnal descent/ascent are directly comparable to the 1999 case despite the different source hardware and time resolution — the point of a d256-compatible grid. Gray band = no receiver coverage below ∼90 km that day.

CADI Support & Interferometric Echo Extraction

```
from pynasonde.digisonde.cadi \
    import CadiExtractor

df = CadiExtractor.load_CADI_files(
    folders=["tmp/CADI/"],
    product="echoes",    # AoA + O/X
        mode
    n_procs=1,
)
# cols: dphi_12, dphi_13, dphi_23,
#       coherence, mode
```



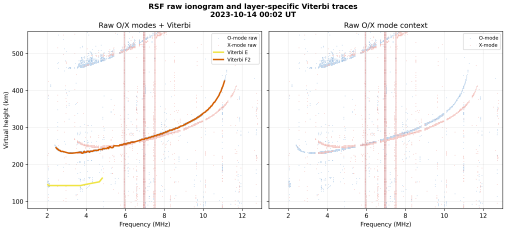
CADI MD2/MD4 reader + `CadiExtractor`: amplitude, phase, cross-receiver baseline phase differences, and coherence per echo.

DIDBase Access & Cross-Format Autoscaling

```
from pynasonde.digisonde.didbase \
    import DidBaseConnector
from pynasonde.digisonde.scaler \
    import scale_dataframe

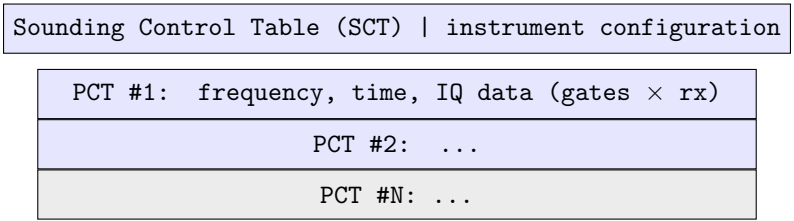
with DidBaseConnector(url, user,
        pwd, role) as db:
    raw = db.fetch_dataframe(
        station="AU930",
        start="2017-05-27",
        end="2017-05-27",
        fmt="MMM")

# same call for RSF, SBF, MMM, CADI
scaled, _ = scale_dataframe(
    raw, mode="O", method="viterbi")
```



One autoscaling code path (schema auto-detection + Viterbi trace tracking) covers RSF, SBF, MMM, and CADI echo tables.

VIPIR RIQ File Structure




```python
from pynasonde.vipir.riq.parsers.read_riq import RiqDataset,
    VIPIR_VERSION_MAP

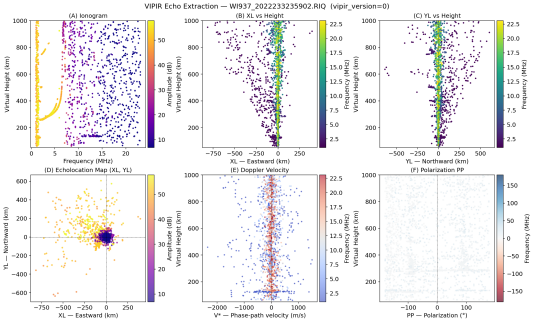
riq = RiqDataset.create_from_file(
    "data/WI937_20220821.RIQ",
    vipir_config=VIPIR_VERSION_MAP.configs[1],    # older WI937 format
)
print(f"Gates  : {riq.sct.timing.gate_count}")    # 960
print(f"Freqs  : {riq.sct.frequency.base_steps}")
```

VIPIR — Echo Extraction

```
from pynasonde.vipir.riq.echo \
    import EchoExtractor

extractor = EchoExtractor(
  sct=riq.sct,
  pulsets=riq.pulsets,
  snr_threshold_db=3.0,
  min_rx_for_direction=3,
).extract()

df = extractor.to_dataframe()
```



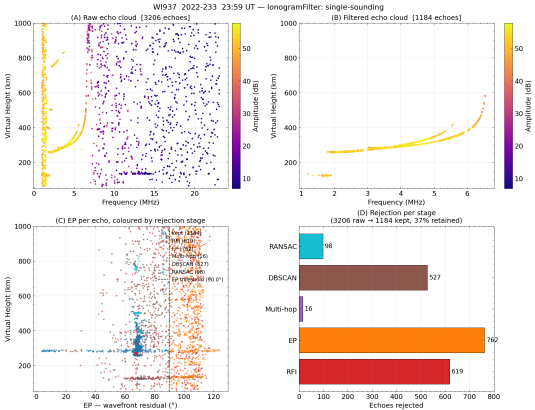
Six-panel diagnostic: ionogram, XL/YL direction cosines, Doppler V*, PP polarization.

VIPIR — Ionogram Filter

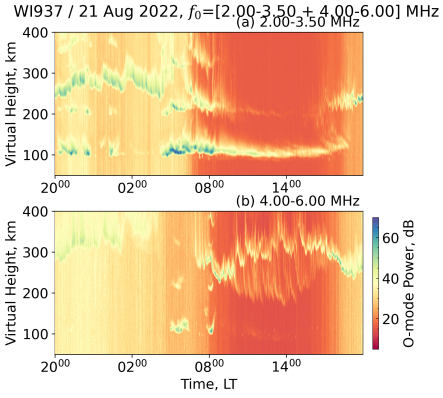
```
from pynasonde.vipir.riq.parsers.\
    filter import IonogramFilter

filtered = IonogramFilter(
  snr_threshold_db=6.0,
  ep_threshold_deg=30.0,
  use_dbscan=True,
  use_ransac=True,
).fit(df)
```

6 stages: RFI → EP → multi-hop → DBSCAN → RANSAC → temporal coherence



VIPIR — FTI (Frequency–Time Interval)



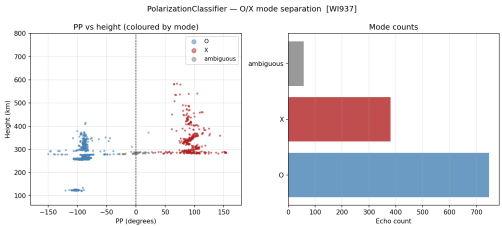
O-mode power FTI panels from a full day of WI937 NGI ionograms.

O/X Mode Separation

```
from pynasonde.vipir.analysis \
    import PolarizationClassifier

clf = PolarizationClassifier(
  o_mode_sign=-1,        # N.
      hemisphere
  pp_ambiguous_threshold_deg=20,
)
pol = clf.fit(filtered_df)
print(pol.summary())
# O=412  X=388  ambiguous=54

o_df = pol.o_mode_df()
```



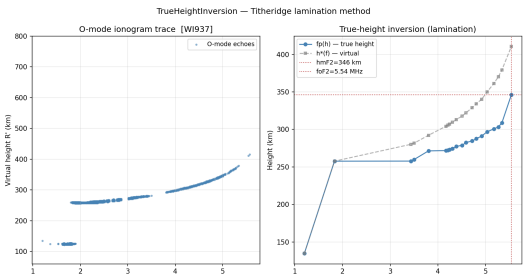
True Height Inversion → N(h) [In Development]

```
from pynasonde.vipir.analysis \
    import TrueHeightInversion

inv = TrueHeightInversion()
edp = inv.fit_from_df(o_df)

print(edp.summary())
# foF2=8.12 MHz
# hmF2=287.4 km
# NmF2=8.2e5 cm^-3

edp.plot()
```

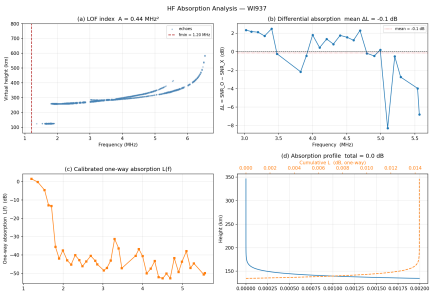


In development — implemented in the codebase but not yet validated/released.

HF Absorption [In Development]

```python
from pynasonde.vipir.analysis \
    import AbsorptionAnalyzer
import numpy as np

aa = AbsorptionAnalyzer()
# 1. LOF index (no calibration)
lof = aa.lof_absorption(df)
# 2. Differential O/X
diff = aa.differential_absorption(
    pol.annotated_df)
# 3. Height-resolved profile
prof = aa.absorption_profile(
  edp,
  nu_hz=lambda h: 1e6*np.exp(-h/8))
```



In development — implemented in the codebase but not yet validated/released.

Sporadic-E — Why Higher Resolution? [In Development]

This entire section (Capon imaging) is implemented in the codebase but not yet a validated/released feature.

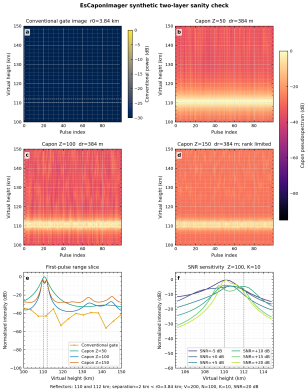
- Es: thin metallic-ion layer at 90–130 km (<1 km thick)
- VIPIR native gate $r_0 = 1.499$ km — **cannot resolve thin layers**
- **Capon cross-spectrum imaging** (Liu et al. 2023): **10× finer resolution**, no extra hardware
- Key: FFT of range profile as spatial frequency array; minimum-variance beamforming in *range space*

Effective resolution

$$\Delta r = \frac{r_0}{K}$$

Instrument	r_0	Δr (K=10)
WISS	3.84 km	384 m
VIPIR	1.499 km	150 m

Es Imaging Sanity Check — Liu et al. Fig. 1



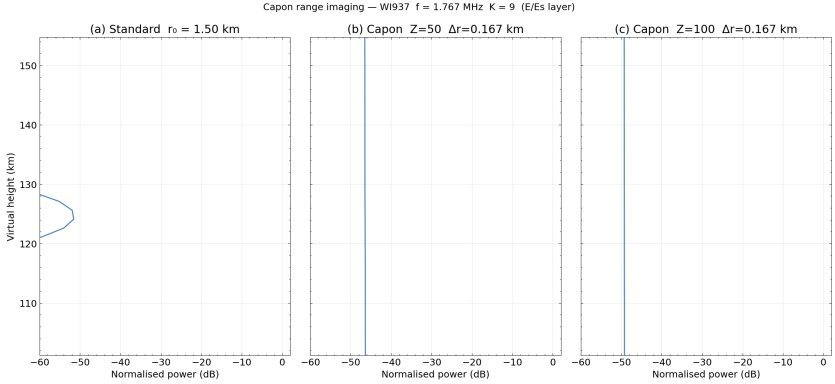
Synthetic two-layer test (110 + 112 km). $Z = 50$: unresolved. $Z = 100$: peaks at 109.8 & 111.7 km ($<$0.3 km error). $Z = 150$: rank-deficient (matches paper Fig. 1d).

Es Imaging — Single File

```
from pynasonde.vipir.analysis import EsCaponImager

imager = EsCaponImager(
    n_subbands=100,          # Z  (Z <= (V+1)/2 recommended)
    resolution_factor=10,    # K  (no singularity constraint)
    gate_spacing_km=1.499,   # VIPIR native gate r0
    gate_start_km=90.0,
)
result = imager.fit(iq_cube)    # (pulses, gates) or (pulses, gates, rx)
print(result.summary())
# EsImagingResult: Z=100  K=10  r0=1.499 km -> Dr=0.150 km
result.plot(snapshot=0)
```

Es Imaging — Single File Result



Es Imaging — Multi-File A+B+C

When only 4 pulses/file: combine 8 files $\times$ 4 pulses $\times$ 8 Rx $=$ 256 samples:

```
from pynasonde.vipir.analysis import RiqAggregator
import glob

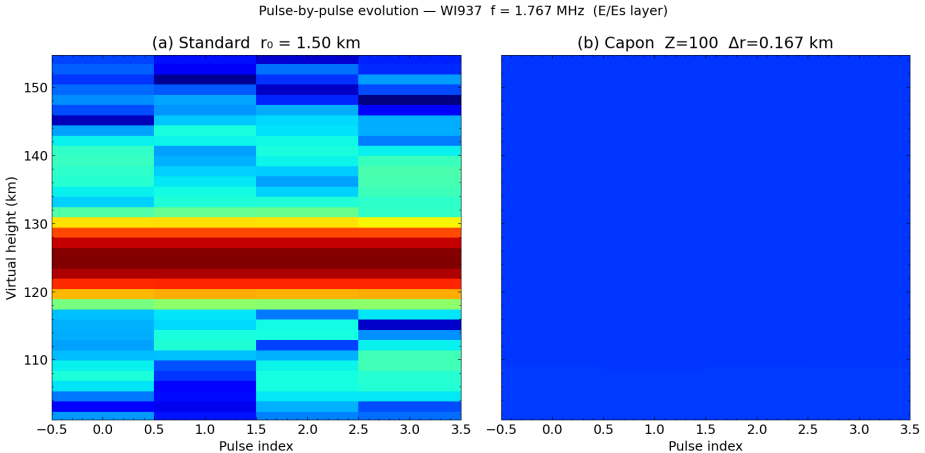
files = sorted(glob.glob("data/20230601_05??.RIQ"))[:8]

agg = RiqAggregator(
    n_subbands=100, resolution_factor=10,
    output_mode="slow_rti",    # one column per file
)
result = agg.fit(files, freq_target_khz=5000.0)
result.plot()
```

SNR improvement

Option A (8 Rx coherent): +9 dB — Option B (4-pulse avg): $\div 4$ var — Option C (8-file stack): $\div 8$ var $\Rightarrow$ $\sim$24 dB total

Es Imaging — Slow RTI (A+B+C)



Slow RTI: one column per file (∼1 min cadence). Each column is a Capon pseudospectrum averaged over Option A+B.

Hands-On Exercises

DIGISONDE

1. Load a `.SAO` file. Print foF2 and hmF2.
2. Build a daily isodensity contour from multiple SAO files.
3. Load a `.DVL` file and plot the Vz time series.
4. Load a `.DFT` file and produce a Doppler waterfall.
5. Split a `.GRM` archive and plot the resulting MMM ionogram.
6. Load CADI files with `product="echoes"`. What fraction is O-mode?
7. Run `scale_dataframe(..., method="viterbi")` on an RSF file, then on an MMM file. Compare foF2.

VIPIR

8. Load a WI937 RIQ file; print gate count and frequency range.
9. Run `IonogramFilter`. How many echoes survive?
10. Separate O/X modes with `PolarizationClassifier`. What fraction is O-mode?
11. Run `IonogramScaler` on the filtered O-mode trace. What is foF2?

* *(In development)* Try `TrueHeightInversion` or `EsCaponImager` and compare against the scaled foF2.

Summary — DIGISONDE I/O (1/3)

DIGISONDE I/O — 9 file formats

- `SaoExtractor` — SAO 4.3/5.0: N(h) + scaled params
- `RsfExtractor` — direction-coded ionogram + directogram
- `SbfExtractor` — SBF binary raw soundings
- `DftExtractor` — Doppler waterfall + spectra
- `DvlExtractor` — drift velocities Vx, Vy, Vz
- `SkyExtractor` — polar sky maps (az/zen)
- `EdpExtractor` — electron density profiles
- `GrmSplitter` **[new]** — auto-splits RSF/SBF/MMM archives
- `CadiExtractor` **[new]** — MD2/MD4, interferometric echoes
- `DidBaseConnector` **[new]** — GIRO DIDBase remote access
- `scale_dataframe()` **[new]** — Viterbi scaling across RSF/SBF/MMM/CADI

Summary — VIPIR I/O (2/3)

VIPIR I/O — 3 file formats

- `RiqDataset` — SCT + PCT loader, `VIPIR_VERSION_MAP`
- `EchoExtractor` **[new]** — 7-param echoes (h, A, V*, EP, PP, XL, YL)
- `IonogramFilter` **[new]** — 6-stage: RFI, EP, multi-hop, DBSCAN, RANSAC, temporal
- `IonogramScaler` **[new]** — foE, foF2, h'F, MUF(3000)
- `PolarizationClassifier` **[new]** — O/X via PP chirality
- `DataSource` — NGI/SDR NetCDF loader + data cube
- `FullPipeline` — end-to-end RIQ $\rightarrow$ N(h) automation

Summary — Analysis Layer (3/3) (In Development)

Not yet validated/released

- `TrueHeightInversion` — Titheridge/POLAN $\rightarrow$ N(h)
- `AbsorptionAnalyzer` — LOF / O-X diff / $\kappa(z)$ profile
- `SpreadFAnalyzer` — range / freq / mixed classification
- `IrregularityAnalyzer` — $\alpha(h)$, outer scale, anisotropy
- `EsCaponImager` — Capon imaging ($\Delta r = 150$ m)
- `RiqAggregator` — A+B+C multi-file ($\sim$24 dB SNR)

pynasonde.readthedocs.io

```
pip install pynasonde
```

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